Epistemic Uncertainty in LLM Hallucinations

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Abstract. Large language models (LLMs) remain prone to hallucination—generating fluent but factually incorrect outputs. A central hypothesis is that hallucinations coincide with elevated *epistemic uncertainty*: when a model generates tokens outside its reliable knowledge, token-level probabilities become less confident. This paper presents a reproducible pipeline that investigates whether uncertainty signals from a low-cost local surrogate language model (`distilgpt2`) can classify hallucinated versus factual answers in the HaluEval QA benchmark. We implement ten complementary baseline detectors spanning four detection paradigms—lexical, uncertainty-based, semantic, and evidence/retrieval-grounded—alongside four hybrid models that combine these signal families, including a fine-tuned cross-encoder upper bound. An ablation study, structured error analysis, and two external evaluations complement the main benchmark. Results show the fine-tuned cross-encoder achieves the highest accuracy and macro F1 on the HaluEval validation split, while the Evidence-Aware Hybrid attains the highest AUROC; retrieval context yields measurable but modest gains over memory-only uncertainty features, suggesting token-level log-probability signals from a cheap surrogate are weak in isolation but complementary when combined with lexical or semantic cues. However, cross-dataset transfer is poor, and the Qwen-generated evaluation further suggests substantially weaker performance on actual generated answers, though this setting remains noisy due to prompt leakage and automatic labeling.

Keywords: Hallucination detection · Epistemic uncertainty · Large language models · Token log-probabilities · Retrieval-augmented generation · HaluEval

1 Introduction

Large language models (LLMs) remain prone to *hallucination*: fluent but unsupported or false outputs [1]. This matters most in high-stakes settings, where confidence and factuality can diverge sharply. In question answering, an answer may sound polished, specific, and locally coherent while still being factually wrong. As models become more capable, scalable, and conversational [12,13], this gap between fluency and truthfulness becomes harder for users to detect without external verification.

One natural hypothesis is that hallucinations correlate with elevated *epistemic uncertainty*. Following the standard distinction between epistemic and aleatoric uncertainty [3], if a model is generating in regions where its knowledge is weak or poorly grounded, token-level probabilities may become less stable or less confident. This motivates the use of log-probability summaries, entropy, and related white-box signals as potential hallucination indicators. At the same time, modern models are also known to be overconfident [9], so uncertainty is not guaranteed to track factuality cleanly. A useful detector therefore needs to test not only whether uncertainty carries signal, but also whether that signal remains useful once stronger semantic and evidence-based features are available.

This problem sits between two different goals that are often conflated in the literature. One goal is to build a practically robust hallucination detector for arbitrary open-domain model outputs. The other is to understand, in a controlled benchmark setting, which signal families are informative and how they interact. Our project is explicitly closer to the second goal. We use HaluEval QA [2] as the main training and benchmarking environment because it provides paired factual and hallucinated answers with a shared question and knowledge snippet, making it suitable for comparative analysis across multiple detectors. At the same time, earlier work on faithfulness and factuality warns that hallucination is not a single failure type: intrinsic and extrinsic errors can behave differently, and confidence can degrade especially on long-tail knowledge or weakly grounded claims [14,15].

We study this question with a fully reproducible local pipeline built around `distilgpt2` [8]. Our three research questions are: **(RQ1)** whether cheap uncertainty features are predictive at all, **(RQ2)** whether retrieved context improves them, and **(RQ3)** whether uncertainty helps when fused with lexical, semantic, and evidence-based signals. The main contribution is therefore not a claim of universal hallucination detection, but a comparative benchmark: ten baselines, ablations, structured error analysis, and transfer evaluation on both TruthfulQA-derived pairs and Qwen-generated TruthfulQA answers. This framing matters because, as our transfer results later show, strong in-domain performance may still fail to generalise to open-domain generated outputs.

2 Related Work

Prior work motivating this paper falls into three strands. The first is **benchmarking hallucination itself**. HaluEval [2] provides paired factual and hallucinated responses across multiple tasks and is well suited to controlled binary classification. TruthfulQA [16] takes a different angle: it targets questions that invite confident but false answers, making it especially valuable for testing whether in-domain benchmark gains survive transfer to harder open-domain settings. Together, these benchmarks suggest that high apparent accuracy inside one dataset should not be equated with general factual robustness. Related evaluation work such as FActScore [17] makes a similar point from another direction: factual reliability is often better understood at the level of supported claims than through surface plausibility alone.

The second strand is **uncertainty-based introspection**. Kadavath et al. [4] show that language models can, to some degree, reflect their own knowledge boundaries. Guo et al. [9] show more broadly that modern neural models are often miscalibrated, which is important because probability scores may still be informative even when they are not perfectly trustworthy as absolute confidence estimates. Kuhn et al. [5] further argue that naive token-level entropy can be too brittle and propose *semantic entropy*, which groups generations by meaning before measuring uncertainty. We do not implement semantic entropy here, but it serves as an important reference point for interpreting the limits of small surrogate log-probability signals.

The third strand is **evidence-based grounding**. Retrieval-Augmented Generation (RAG) [6] improves factuality by conditioning model responses on external evidence, while SelfCheckGPT [7] uses agreement across multiple sampled outputs as a black-box reliability cue. These approaches are relevant because they shift attention away from raw token probabilities alone and toward whether an answer is supported by evidence or by consistency across generations. Our NLI and context-comparison baselines are inspired by this view.

Our contribution relative to this literature is narrower but practical: rather than proposing a new frontier-scale detector, we compare uncertainty, lexical, semantic, and evidence-based signals inside one local and reproducible pipeline. That design lets us ask not only which method works best on HaluEval, but also which families of signals degrade most severely once the evaluation moves to TruthfulQA and actual model-generated answers.

3 Methodology

The pipeline has four stages: dataset preparation, local LM scoring, baseline training, and hybrid-model training. All stages are orchestrated by `src/main.py`.

3.1 Dataset and Problem Formulation

We use the QA split of HaluEval [2]. Each record provides a question, a knowledge snippet, one factual answer, and one hallucinated answer. We expand each record into two supervised rows, factual ($y = 0$) and hallucinated ($y = 1$), producing balanced train/validation/test JSONL splits via `scripts/prepare_dataset.py`. The main experiments use 15,999 training examples and 2,001 validation examples. This paired construction is useful for controlled comparison, but it may also introduce dataset-specific cues.

3.2 Local Language Model Scoring

Token-level log-probabilities are extracted with `distilgpt2` [8], used as a cheap local surrogate. For each candidate answer we compute mean log-probability, minimum log-probability, predictive entropy, and answer length. Scoring is done in both *memory mode* (question + answer) and *context mode* (knowledge prepended), allowing us to measure whether external evidence changes model confidence.

3.3 Single-Signal Baselines

We compare five single-signal baselines: a TF-IDF lexical SVM, two uncertainty classifiers (`entropy_base` with `distilgpt2` and `entropy_upgraded` with `Qwen/Qwen2.5-1.5B-Instruct`), a semantic SVM built from sentence-transformer interactions, a context-sensitive RAG comparison model based on the confidence shift $\Delta\bar{\ell}$, and an NLI evidence model using `roberta-large-mnli`. Together they represent shallow lexical cues, surrogate uncertainty, semantic compatibility, and evidence alignment.

3.4 Hybrid and Joint Reasoning Models

We also train four joint models: lexical and semantic hybrids that append uncertainty features to the corresponding base representations, an Evidence-Aware Hybrid that combines semantic, uncertainty, context-shift, and NLI signals, and a fine-tuned `distilroberta-base` cross-encoder as an upper-bound supervised baseline.

3.5 Ablation Study Design

Four ablations isolate feature-group contributions: *lexical_ only*, *uncertainty_ only*, *hybrid_ no_ context*, and *hybrid_ with_ context*.

4 Experiments

4.1 Setup

The experiments use `Python 3.11`, `PyTorch 2.4`, `Transformers 4.44`, `scikit-learn 1.5` [10], and dependencies pinned in `requirements.txt`. The classification runs are fully local and reproducible; some heavier generation and scoring steps can additionally use GPU when available. Model weights are downloaded automatically via the Hugging Face Hub. All randomness is controlled by fixed seeds. We report Accuracy, Precision, Recall and macro-averaged F1 on the validation split. A random baseline achieves 50% accuracy on the balanced splits.

4.2 Classification Results

Table 1 summarises performance across all ten methods. The Evidence-Aware Hybrid achieves the highest AUROC (0.9951), while the fine-tuned Cross-Encoder leads on Accuracy and Macro F1 (0.9815). Among single-signal baselines, the Semantic SVM is strongest; both entropy baselines trail the richer semantic and hybrid approaches.

Table 1: Comprehensive hallucination detection results on the HaluEval QA validation split. Best results per metric are in bold.

	Method	Acc.	Macro F1	AUROC
Single-Signal	Lexical SVM	0.9381	0.9380	0.9729
	Entropy Base	0.9426	0.9425	0.9765
	Entropy Upgraded	0.9231	0.9231	0.9654
	Semantic SVM	0.9695	0.9695	0.9870
	NLI Evidence	0.7877	0.7866	0.7943
	RAG Compare Fixed	0.7522	0.7490	0.7723
Hybrid	Lexical Hybrid SVM	0.9530	0.9530	0.9805
	Semantic Hybrid SVM	0.9675	0.9675	0.9878
	Evidence-Aware Hybrid	0.9770	0.9770	**0.9951**
	Cross-Encoder (FT)	**0.9815**	**0.9815**	0.9895

Table 2: Ablation study: marginal contribution of each feature group on the HaluEval QA validation split (linear SVM classifier, $n = 2{,}002$ validation rows).

Setting	Features	Acc.	Macro F1	AUROC
lexical_only	TF-IDF	0.9391	0.9390	0.9703
uncertainty_only	Log-probability features	0.7338	0.7152	0.9740
hybrid_no_context	TF-IDF + log-probability features	0.9446	0.9446	0.9800
hybrid_with_context	TF-IDF + log-probability + Δ	**0.9735**	**0.9735**	**0.9919**

4.3 Ablation Results

Table 2 reports the ablation study for the lexical-uncertainty model family on the HaluEval QA validation split. Lexical features already provide a strong baseline (Macro F1 = 0.9390). Uncertainty features alone are substantially weaker under the default threshold (Macro F1 = 0.7152), although their AUROC remains high (0.9740), indicating useful ranking signal even in isolation. Adding the context-delta features produces the strongest ablation result overall, directly supporting RQ2.

4.4 Transfer Evaluation

To test external validity, we evaluated the HaluEval-trained detectors on two out-of-domain settings: a TruthfulQA-derived binary evaluation set (1,634 rows) and a harder set of 817 `Qwen/Qwen2.5-1.5B-Instruct` generations from TruthfulQA questions. Table 3 shows the main pattern. High HaluEval accuracy does not transfer reliably. The Cross-Encoder drops from 0.9815 to 0.5477 on TruthfulQA pairs and 0.4712 on Qwen-generated answers; the Semantic SVM drops from 0.9695 to 0.5208 and 0.4749; the Entropy Base model falls from 0.9426 to 0.5024

Table 3: Transfer performance from HaluEval QA training to two out-of-domain evaluation sets.

Method	HaluEval	TruthfulQA	TQA AUROC	Qwen-Gen	QG AUROC
Cross-Encoder	0.9815	0.5477	0.5764	0.4712	0.4924
Entropy Base	0.9426	0.5024	0.4863	0.4847	0.5062
Semantic SVM	0.9695	0.5208	0.4750	0.4749	0.4344
Evidence-Aware	0.9770	0.5967	0.8665	0.4847	0.4832
NLI Evidence	0.7877	0.9694	0.9943	0.5129	0.7034
RAG Compare	0.7522	0.8635	0.9514	0.4847	0.5506

and 0.4847. Evidence-grounded methods transfer better on benchmark-written TruthfulQA answers, especially NLI Evidence (0.9694 Accuracy, 0.9943 AUROC), but they also weaken sharply on the Qwen-generated set. We interpret that second result more cautiously than the canonical TruthfulQA transfer table, because a post-hoc audit revealed prompt leakage and weak automatic NLI labels in part of the generated evaluation set. Even with that caveat, the main qualitative pattern remains useful: the pipeline is no longer evaluated only inside HaluEval, and the transfer gap is explicit rather than assumed.

5 Error Analysis

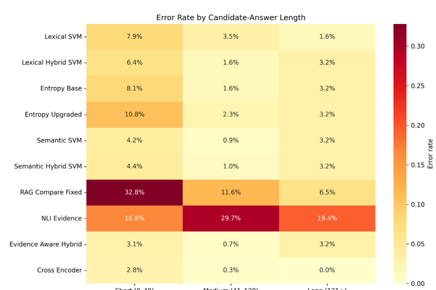
We analyse errors by answer length and decision-margin confidence because raw aggregate metrics hide distinct failure modes. Figure 1 shows that lexical, uncertainty, semantic, and evidence-based systems break down under different conditions even when their headline scores are all high.

Short answers are the hardest subgroup. Across nearly all baselines, error rates peak in the shortest answer bucket and drop as answers become longer. This is intuitive once we inspect the feature families. Short responses contain fewer informative n-grams, fewer tokens over which log-probability statistics can stabilize, and less semantic structure for interaction models to use. Many HaluEval hallucinations are also short and stylistically plausible, so they can look like normal factual answers unless the model has access to a strong evidence signal. This is consistent with the broader phenomenon of high-confidence hallucination, where wrong answers can remain locally fluent and confident despite being factually unsupported [11]. In this regime, even small wording choices can dominate the decision boundary.

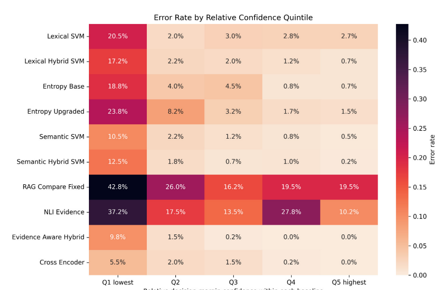
This finding turns a reviewer criticism into a concrete modeling diagnosis. We tested a length-aware extension of the Evidence-Aware Hybrid using separate short/long experts with a 40-token threshold. However, it did not improve results (Accuracy 0.9770 $\rightarrow$ 0.9760; AUROC 0.9951 $\rightarrow$ 0.9950). The likely reason is that HaluEval is overwhelmingly short-form under this definition, so the split does not create two genuinely different learning regimes. In other words, the weakness is real, but a naive bucketed solution is not enough. Better short-answer handling will likely require stronger entity matching, tighter evidence comparison, or calibration strategies designed specifically for brief and underspecified claims.

Evidence-only signals are more overconfident. The confidence stratification also reveals an important contrast between model families. Lexical and semantic models concentrate most of their errors in the lowest confidence region; once they become even moderately confident, their error rates fall quickly. By contrast, evidence-comparison baselines such as RAG Compare and, to a lesser extent, NLI Evidence retain noticeable errors even at higher decision margins. This suggests that the raw confidence shift caused by adding context is not a clean factuality score by itself. A large or small $\Delta\bar{\ell}$ may reflect stylistic compatibility with the added context, surface overlap, or quirks of the surrogate language model rather than true grounding.

Hybrid models fail more gracefully. The best joint systems reduce exactly these brittle behaviors. The Evidence-Aware Hybrid combines semantic interaction, NLI support, and uncertainty summaries, and its error profile is much flatter in the high-confidence region than the evidence-only baselines. This supports the broader interpretation of the paper: uncertainty and evidence signals are useful, but they are strongest when they act as complementary constraints rather than as standalone decision rules. The error analysis therefore reinforces the main quantitative results. The problem is not that uncertainty is useless; it is that isolated uncertainty and isolated evidence shifts remain too fragile on the most ambiguous, short, and plausible answers.



(a) Stratification by answer length.



(b) Stratification by confidence.

Fig. 1: Error rate analyses across baseline models.

6 Conclusion

Interpretation. Our results support a narrower claim than a generic "hallucination detector" story. Token-level log-probability features from `distilgpt2` carry real signal, but they are not the strongest standalone family and work best when combined with semantic or evidence-aware features. Replacing `distilgpt2` with `Qwen2.5-1.5B-Instruct` did not help: the upgraded entropy baseline drops from 0.9425 to 0.9231 Macro F1, suggesting that stronger next-token modeling does not automatically yield a more useful factuality signal. The ablation and hybrid results point to the same practical lesson: uncertainty is complementary, not sufficient, and the change in model confidence after adding evidence is more informative than raw confidence alone.

Limitations and Future Work. The main limitation remains external validity. Although we now evaluate on both TruthfulQA-derived pairs and Qwen-generated TruthfulQA answers, all supervised training is still done on HaluEval QA, whose paired construction can reward dataset-specific artifacts. The uncertainty signal is also taken from a surrogate model rather than from the original generator, so it should be interpreted as a cheap proxy rather than a direct measure of the source model's epistemic state. In addition, the Qwen-generated TruthfulQA evaluation set was labeled automatically using NLI, and our audit found noticeable prompt leakage and many low-margin label decisions; that generated-output experiment is therefore best read as indicative rather than fully clean ground truth. Short answers also remain a genuine weakness: we tested a simple length-aware expert split, but it did not improve results because HaluEval is overwhelmingly short-form. The next step is therefore not merely "add a length feature," but to move toward stronger uncertainty families such as semantic entropy [5] or self-consistency, train or calibrate on more realistic open-domain generations, and design short-answer handling that relies more on entity and evidence matching than on coarse length buckets.

Conclusion. We presented a reproducible pipeline for comparing uncertainty, semantic, and evidence-based hallucination detectors on HaluEval QA. Within the benchmark, hybrid and cross-attentive models perform extremely well. Outside it, however, performance drops sharply, and the Qwen-generated evaluation further suggests that real model outputs are harder than benchmark-written answers, even though that generated set still contains label noise and prompt artifacts. The main contribution of the project is therefore methodological and diagnostic: it shows where cheap uncertainty signals help, where they fail, and why open-domain hallucination detection still requires stronger generation-aware and evidence-aware modeling.

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